

The Five API Aspects a TPM Must Read

Day 3 handout. You won't write the OpenAPI spec line by line — but you must be able to review one and ask sharp questions. These are the five aspects to check in any API contract.

ASPECT	QUESTION TO ANSWER
Resources	What nouns does the API expose? Are they consistent?
Methods	What verbs apply? Are they used consistently?
Idempotency	Can a client safely retry? How? (Idempotency key? Natural idempotence?)
Versioning	How does the API evolve without breaking existing callers?
Errors	What does a failure look like? Do error codes carry actionable info?

REST vs SOAP — the default and the exception

- **REST (often with JSON)** is the right default for most new B2B SaaS features.
- **SOAP** (XML over HTTP, with a WSDL contract) appears at integration boundaries with legacy partners. Consider it only when:

- The integration partner only speaks SOAP (banks, government, EDI-adjacent systems).
- WS-Security or WS-ReliableMessaging is contractually required.
- The partner has invested heavily in SOAP tooling and rewriting them is impractical.

Default to REST. Justify any SOAP touchpoint against a **named partner requirement** — "just in case" SOAP support is gold-plating.